Mapping the Quantum Computing Landscape: Growth, Collaboration, and Thematic Convergence

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Supporting Information

Table S1. The search query was constructed in Scopus to comprehensively capture the quantum computing research landscape across seven thematic modules: (1) quantum computing and quantum-mechanical models, (2) quantum information, communication, and networking, (3) quantum cryptography, (4) quantum sensing and control, (5) quantum metrology, tomography, clocks, gravimeters, and imaging, (6) quantum hardware systems, and (7) quantum simulation, algorithms, and software. Exclusion filters were applied to remove documents primarily related to materials science and condensed matter physics. Proximity operators, controlled vocabularies, and exclusion clauses were incorporated to optimize precision and recall. Document types were restricted to articles (ar), reviews (re), book chapters (ch), books (bk), and conference papers (cp), following Scopus document type codes.

Query Block	Boolean Query	Rationale
General Exclusions (Pre-filtering applied to all blocks)	*TITLE-ABS-KEY(NOT ("neutron*" OR "scattering" OR "polycrystalline*" OR "thin film*" OR "boundary condition*" OR "theory of chemical-reaction ")) AND DOCTYPE(ar OR re OR ch OR bk OR cp)*	A set of exclusions (neutron*, scattering, polycrystalline*, thin film*, boundary condition*, theory of chemical-reaction) filters out materials science, condensed matter physics, and chemistry papers that include the term "quantum" but are unrelated to quantum information or quantum computing. The search is further restricted to standard research outputs by limiting document types to articles (ar), reviews (re), book chapters (ch), books (bk), and conference papers (cp), following Scopus document type codes.
1. Quantum Computing & Quantum-Mechanical Models	*TITLE-ABS-KEY((("quantum comput*" AND ("computer*" OR "Turing machine*" OR "tensor network*") OR ("quant* mec* Hamilton*" AND ("comput*" OR "Turin* machin*")) OR "quantum information" OR ("quantum algorithm*" AND ("computer*" OR "Turing machine*"))OR ("quantum circuit*" AND ("computer*" OR "Turing machine*")) OR "quantum error correction" OR ("variational quantum eig*" AND ("computer*" OR "Turing machine*")) OR "quantum anneal*" OR "quantum machine learning" OR "quantum Turing machine*" OR "qubit*" OR ("quantum mechanical model*" AND ("computer*" OR*	"quantum comput*" paired with "computer*", "Turing machine*", or "tensor network*" captures both theoretical and implementation-level quantum computation research. "quant* mec* Hamilton*" combined with "comput*" or "Turin* machin*" locates works on quantum mechanical Hamiltonians in computational contexts. "quantum information" is included to cover information-processing studies without explicit computation qualifiers. Terms such as "quantum algorithm*", "quantum circuit*", "quantum error correction", "variational

	"Turing machine")) OR "microscopic quantum mechanical Hamiltonian model")*	quantum eig*", "quantum anneal*", and "quantum machine learning" retrieve key algorithmic and hardware developments. "quantum Turing machine*", "qubit*", and "quantum mechanical model*" capture foundational models of quantum computation.
2. Quantum Information, Communication & Networking	*TITLE-ABS-KEY("quantum information*" OR "von Neumann mutual information" OR "quantum mutual information" OR "quantum fisher information") OR TITLE-ABS-KEY(quantum W/2 technolog*) OR TITLE-ABS-KEY("quantum communication*" OR "quantum network*" OR "quantum optical communication" OR "quantum state transmission*" OR (("quantum memor*" OR "quantum storage*") W/5 photon*) OR "quantum repeater*" OR "quantum internet" OR ("quantum teleport*" AND (qubit* OR "quantum bit*" OR entangle*)))*	"quantum information*" and exact phrases like "von Neumann mutual information", "quantum mutual information", and "quantum fisher information" include key works in quantum information theory. quantum W/2 technolog* retrieves variants of "quantum technology" while excluding terms like "quantum dot display technology." Terms like "quantum communication*", "quantum network*", "quantum optical communication", and "quantum state transmission*" target quantum information transfer. (("quantum memor*" OR "quantum storage*") W/5 photon*) captures optical quantum memory and storage studies. "quantum repeater*" and "quantum internet" address quantum communication infrastructures. ("quantum teleport*" AND (qubit* OR "quantum bit*" OR entangle*)) focuses on teleportation implementations tied to qubits or entanglement, excluding purely theoretical treatments.
3. Quantum Cryptography	*TITLE-ABS-KEY("quantum crypto*" OR pqcrypto* OR "quantum key distribution" OR "quantum encrypt*" OR (("quantum secur*" OR*	"quantum crypto*" retrieves quantum cryptography and related fields.

	"quantum secre") AND NOT ("quantum secreted" OR "quantum secretion")))*	pqcrypto* captures post-quantum cryptography literature. "quantum key distribution" and "quantum encrypt*" retrieve key QKD and encryption studies. The exclusion NOT ("quantum secreted" OR "quantum secretion") removes biomedical uses of "secretion."
4. Quantum Sensing and Control	*TITLE-ABS-KEY("quantum control*" OR "control* of quantum" OR "control over quantum" OR "quantum optimal control" OR "quantum state control" OR "control* quantum" OR "control* the quantum" OR "quantum coherent control") OR TITLE-ABS-KEY(quantum W/1 sensing OR quantum W/1 sensor*)*	quantum W/1 sensing and quantum W/1 sensor* retrieve quantum-enhanced sensing and measurement studies. Exact phrases ensure coverage of terms like "quantum optimal control", "quantum coherent control", "quantum state control" while excluding irrelevant combinations (e.g., "quantum path control" in chemistry).
5. Quantum Metrology, Tomography, Clocks, Gravimeters & Imaging	*TITLE-ABS-KEY("quantum imag*" OR "ghost imag*") OR TITLE-ABS-KEY(quantum W/10 metrology OR quantum W/1 tomograph* OR "atomic clock*" OR "ion clock*" OR "quantum clock*" OR "quantum gravimeter*")*	quantum W/10 metrology captures long phrases such as "quantum Hall devices for primary resistance metrology." quantum W/1 tomograph* retrieves process and state tomography while excluding classical tomography noise. "atomic clock*", "ion clock*", "quantum clock*", "quantum gravimeter*" target quantum precision measurement devices. "quantum imag*" captures quantum imaging studies. "ghost imag*" specifically includes ghost imaging.
6. Quantum Hardware Systems	*TITLE-ABS-KEY("quantum hardware" OR "quantum device*" OR "quantum circuit" OR "quantum processor*" OR "quantum register*")*	Captures research related to hardware realization of quantum computing platforms, including device design, processors, circuits, and register architectures.

7. Quantum Simulation, Algorithms, and Software	*TITLE-ABS-KEY(("quantum simulat*" AND (qubit* OR "quantum bit*" OR "quantum comput*"))OR "quantum simulator*")) OR TITLE-ABS-KEY("quantum algorithm*") OR TITLE-ABS-KEY("quantum software" OR "quantum cod*" OR "quantum program*")*	"quantum algorithm*" retrieves algorithmic advances (e.g., Shor, Grover, VQE). "quantum software", "quantum cod*", "quantum program*" capture software frameworks, compilers, and programming languages for quantum computing. ("quantum simulat*" AND (qubit* OR "quantum bit*" OR "quantum comput*")) ensures selection of true quantum simulations rather than classical simulations of quantum phenomena. "quantum simulator*" includes generic simulation platform developments.

Table S2. Mathematical definitions of network metrics and centrality measures used in the analysis.

Metric	Purpose	Formula
Degree Centrality	Measures the local influence of a node via number of direct connections.	$k_i = \sum_{j=1}^{N} A_{ij}$, with A_{ij} being the adjacency matrix
Density	Measure the fraction of possible connections that exist, classifying networks as sparse/dense to guide analysis, storage, and algorithms.	$\frac{\sum_i^N k_i}{N(N-1)}$
Shortest path length and small-world	Quantifies avg. node separation; small-world (low + high clustering) enables rapid global info flow in sparse, clustered real-world networks.	$< l > = \frac{2\sum_{i,j} l_{i,j}}{N(N-1)}$, with $l_{i,j}$ the shortest-path length between nodes *i* and *j*
Scale free: (2$< \gamma < 3$)	Classify networks as hub-driven (robust to random fails, fragile to attacks) vs. uniform, guiding modeling, resilience analysis, and optimization.	$\gamma = 1 + N(\sum_i \ln (\frac{k_i}{\min (k_i)}))^{-1}$
Betweenness Centrality	Identifies nodes acting as bridges or key connectors within different parts of the network.	$b_i = \sum_{h \neq j \neq i} \frac{\sigma_{hj}(i)}{\sigma_{hj}}$
Assortivity	Quantifies if high-degree nodes link to similar high-degree peers (positive: communities) or peripherals (negative: resilience), predicting dynamics like epidemics and info spread.	$k_{nn}(i) = \frac{1}{k_i} \sum_{j \in N} k_j$
Clustering coefficient	Measures local triangle formation around nodes (vs. degree); high values signal dense communities/cliques vs. random sparsity	$\bar{c} = \frac{1}{N} \sum_i \frac{2T_i}{k_i(k_i-1)}$, with T_i being a triangle formed between 3 nodes
Eigenvector Centrality	Captures global prestige: influence of a node via its	$A\ x\ =\ \lambda\ x$,with *A* being the adjacency matrix

	connection to already influential neighbors.	
PageRank	Ranks nodes (e.g., documents) via recursive importance, mimicking citation flow.	$R_t = \frac{\alpha}{N} + (1-\alpha)\sum_{j \in pred(i)} \frac{R_{t-1}(j)}{k_{out}(j)}$ where $\alpha = 0.15$, and the initial value $R_0 = {}^{1}/_{N}$